

Xiaochuan Li

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EDUCATION

M.Sc., Human and AI

2025 — 2027

University of Technology Nuremberg (UTN)

Focus: language, reasoning, and AI — testing whether models use the linguistic structures that license inference, or surface cues that merely correlate.

German language program

2024 — 2025

University of Marburg · intensive German for graduate study

B.A., Philosophy

2020 — 2024

Shenzhen University · College of Humanities

Senior thesis: *On Leibniz's Logical Theory* (advisor: Zang Yong)

Coursework: logic, history of philosophy, analytic philosophy, mathematical logic; electives in cognitive psychology and linguistics. This background shapes my focus on reasons, causality, and what makes an inference licensed.

RESEARCH

Language structure as a scaffold for reasoning

Thesis program · three method contributions

Certify whether models use the linguistic structures that license inference (Stable Is Not Grounded), localize where that structure is used (Isotrace), and quantify the inferential yield it produces (Causal Inferential Yield) — using causal-interpretability tools.

Typological Grounding

Application frontier

Where the three methods meet linguistic typology — the same inferential content re-encoded by each grammar; a test of structure versus surface.

PAPERS IN PROGRESS

Stable Is Not Grounded. Paper 1: binary certification for whether the licensing structure is load-bearing.

Isotrace. Paper 2: behavioral path localization via path-encoded fictional names.

Causal Inferential Yield (CIY). Paper 3: quantitative grading over certified or controlled causal-structure use.

PROJECTS

Causal Direction

personal · pilot complete

A completed mechanistic pilot on a real LLM (Gemma-2-2B). Linear probes, activation steering, and layer-band ablation show the model holds a representation of causal direction — which event is the cause — that is linearly decodable (cross-domain peak ≈ 0.94), causally efficacious under steering, and abstract enough to transfer to fictional, zero-frequency relations, yet redundantly distributed across layers (ablation cannot cleanly remove it). End-to-end interpretability pipeline designed, run, and analyzed on an HPC GPU cluster.

HanGL — surface Hangul, latent Hanja

collaborative

The structure-versus-surface question run on real LLMs through a natural phenomenon: Korean Sino-Korean words, where Hangul shares no characters with its Hanja origin yet matches it in meaning. A wrong-Hanja intervention shows latent grounding scales with model capability — strong models neither need the cue nor are misled by it; a weak one is both. Pilot complete; scaling up.

Latent Control States

collaborative

Does prompt framing — urgency, authority, epistemic positioning — only change an LLM's outputs, or shift a causally relevant latent state? A controlled two-agent data pipeline is built; linear probing and activation patching come next. A collaborative project carried out with, and advised by, a senior researcher at DFKI (the German Research Center for Artificial Intelligence).

Cross-lingual ProtoBias

collaborative

An applied multimodal stress test: does a vision-language model track semantic content, or language- and culture-specific prototype shortcuts? Extends a forthcoming paper from an established NLP research group with a cross-lingual evaluation, now at its second iteration.

MiniCausalLang

personal · workbench

A language-to-causal-graph workbench that makes grammatical structure, the inference it licenses, and interventions on it observable — the shared substrate the certify / localize / quantify methods are built and tested on.

SKILLS

Consolidating the technical stack my research program needs — controlled linguistic data generation, transformer-based model evaluation, intervention-based causal testing, and reproducible experiment pipelines.

Programming & data

Python (NumPy, pandas, scikit-learn), Git, LaTeX.

ML / NLP tooling

Building working proficiency with PyTorch and HuggingFace Transformers — model inference, dataset pipelines, and LoRA fine-tuning.

Causal & interpretability methods

Controlled do-interventions, causal graphs, and shortcut-severing audits; developing behavioral path-tracing.

Linguistic evaluation

Designing controlled stimuli for morphology, agreement, dependency structure, and word order — drawing on grammar-sensitive work across Chinese, German, English, and Japanese.

LANGUAGES

Mandarin Chinese (native) · German (C1) · English (C1) — spanning isolating, fusional, and analytic types. Learning Japanese, to extend the range with an agglutinative, head-final system — the typological span my linguistic work draws on.

INTERESTS

Language structure as a scaffold for reasoning in humans and AI; causal tests of whether models use genuine linguistic structure or merely its surface. The boundary between behavioral and mechanistic methods. Philosophy of measurement and of language in ML. World-model approaches (LeCun, Pearl) and the limits of pure-LLM paradigms.